



05_project / AI / Edge hardware

Off-Grid, AI-Driven Farming

One acre, zero dependency. Two plots 2,256 km apart, a solar node that runs on 0.16 Wh a day, and a local model that plans tomorrow morning on a 4,000 m² field. No cloud anywhere in the daily loop.

Field node

Solar, sealed, 0.16 Wh/day. Sized for a Baltic December.

Data spine

Twenty years of regional weather, provenance kept.

Scheduler

A local model proposes. Tested Python disposes.

// the problem

Smart farming assumes a grid that is not there

Most smart farming projects assume mains power, a cloud connection and a budget for commercial sensor kits. This one starts from the opposite constraints: no grid, no runtime network, about €175 of parts, and a decision that still has to be right on the day it is made.

Those are the constraints of any real edge deployment. A remote asset, a link that drops in exactly the weather you built the thing to observe, a model that has to run somewhere cheap, and an answer that gets acted on before anyone has time to check it.

// what it is

One acre, zero dependency

Two plots, 2,256 km apart, farmed in opposite halves of the year: **Prora** on the German Baltic coast through summer, **Castelo Branco** in central Portugal through winter. A solar-powered field module, the Birdbox, measures soil and local weather.

A laptop-side Python stack merges those readings with twenty years of regional weather history from the Open-Meteo historical archive API (2006-2026), forecasts its own solar budget, computes the Thun day type from first principles, and hands the lot to a locally running model as the day's plan: which crop, in which bed, in which window, on a small field of 4,000 m².

How much of a small farm's daily decision-making can genuinely off-grid AI handle end to end, from sensor to scheduler? The repository is built to answer that, not to assume it.

// the purpose

Who it is for

To give farmers, hobby growers and families something to hold on to, so they can grow organically at small and large scale, responsibly and successfully, on modern technique and age-old concepts that protect the earth.

// three commitments

What the project refuses to fudge

Off-grid

No cloud model in the daily loop

The scheduler runs on the laptop against a local model. The network pulls public history during setup and never becomes a runtime dependency of a decision.

Provenance

Every number carries its error bar

A €30 sensor that claims ± 0.3 pH does not get to state pH to one decimal. Raw readings stay raw, calibration applies on read, and each channel has a defensible uncertainty before it reaches the model.

Falsifiable

The calendar is a hypothesis, not a rule

The evidence does not support the biodynamic trigon effect. So it is computed correctly, logged as a covariate, and tested against a pre-registered analysis. A null result is a valid output, possibly the most valuable one.

// the data

What each source can and cannot say

Open-Meteo archive

Twenty years of hourly regional weather, 2006-2026, identical treatment at both plots. Reanalysis at grid scale, not the Birdbox's microclimate.

DWD Arkona

Measured global radiation 33 km from the northern plot, daily since 1981. Castelo Branco has no equivalent, so the two plots do not carry equal evidence weight.

Eurostat

Regional yield figures, but coarse: annual, per hectare, per region, and for Germany only at NUTS 1. A plausibility check on arable crops, never a per-bed prediction.

Thun calendar

Computed from the ephemeris, sidereal not tropical. Logged beside every planting as a covariate under test.

Own harvest logs

No public archive reports vegetable yield per square metre for either region. Season one is therefore its own baseline, and that is stated rather than hidden.